Assignment behavior of the selected training seeds

v1_matched

MLP · s1 — RNN single-partner · s5
RNN · s5

v2

MLP · s4 — RNN single-partner · s4
RNN · s2

v3

MLP · s4 — RNN single-partner · s5
RNN · s3

Final ego-red assignment (%)

Final ego-blue assignment (%)

Partner on faster task (%)

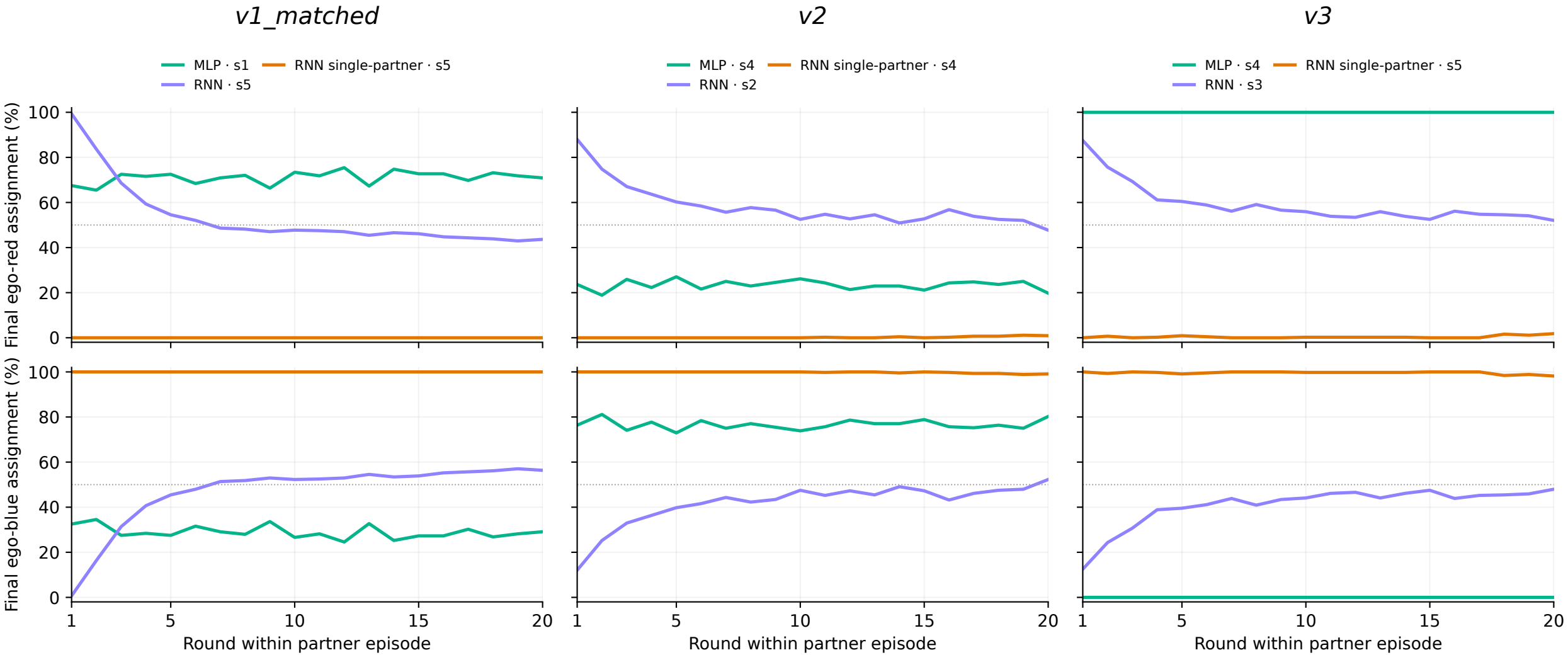
Mean switches / round

Assignments fixed within each round

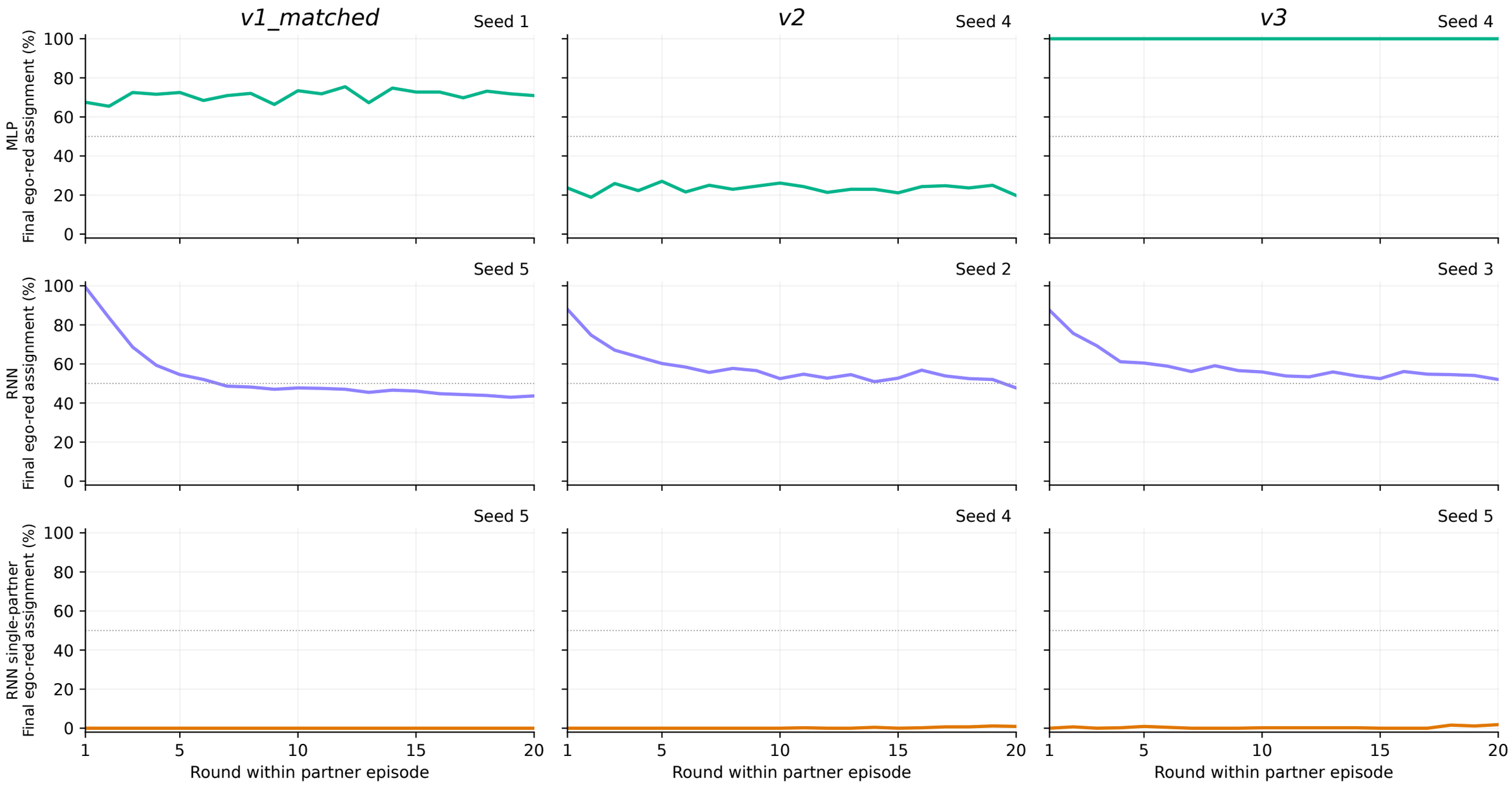
Round within partner episode

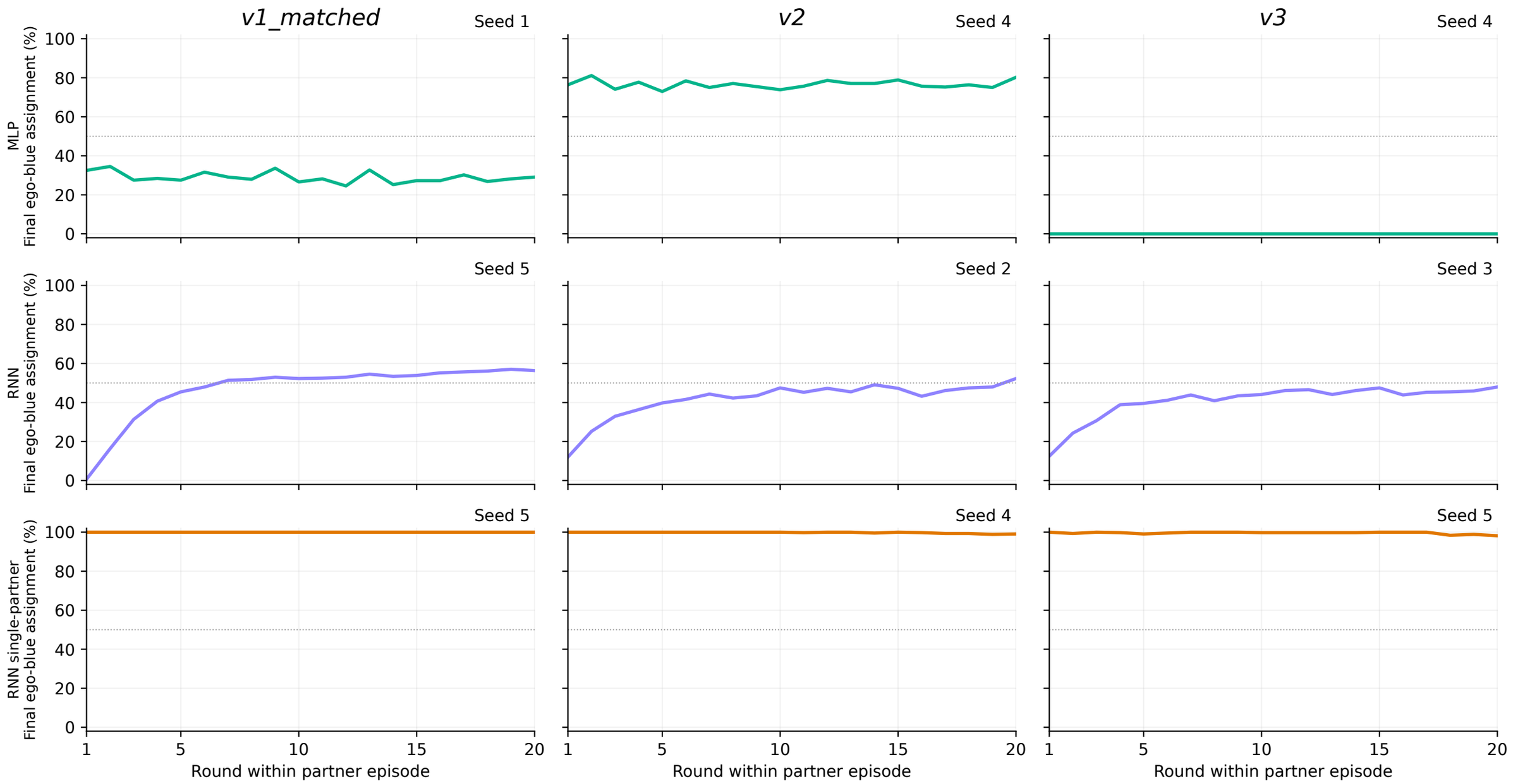
Same policy selections as the best-of-five held-out return figure; selected by final logged training return.
Red = ego red / partner blue; blue = ego blue / partner red. Final assignments average 440 episodes at each round.

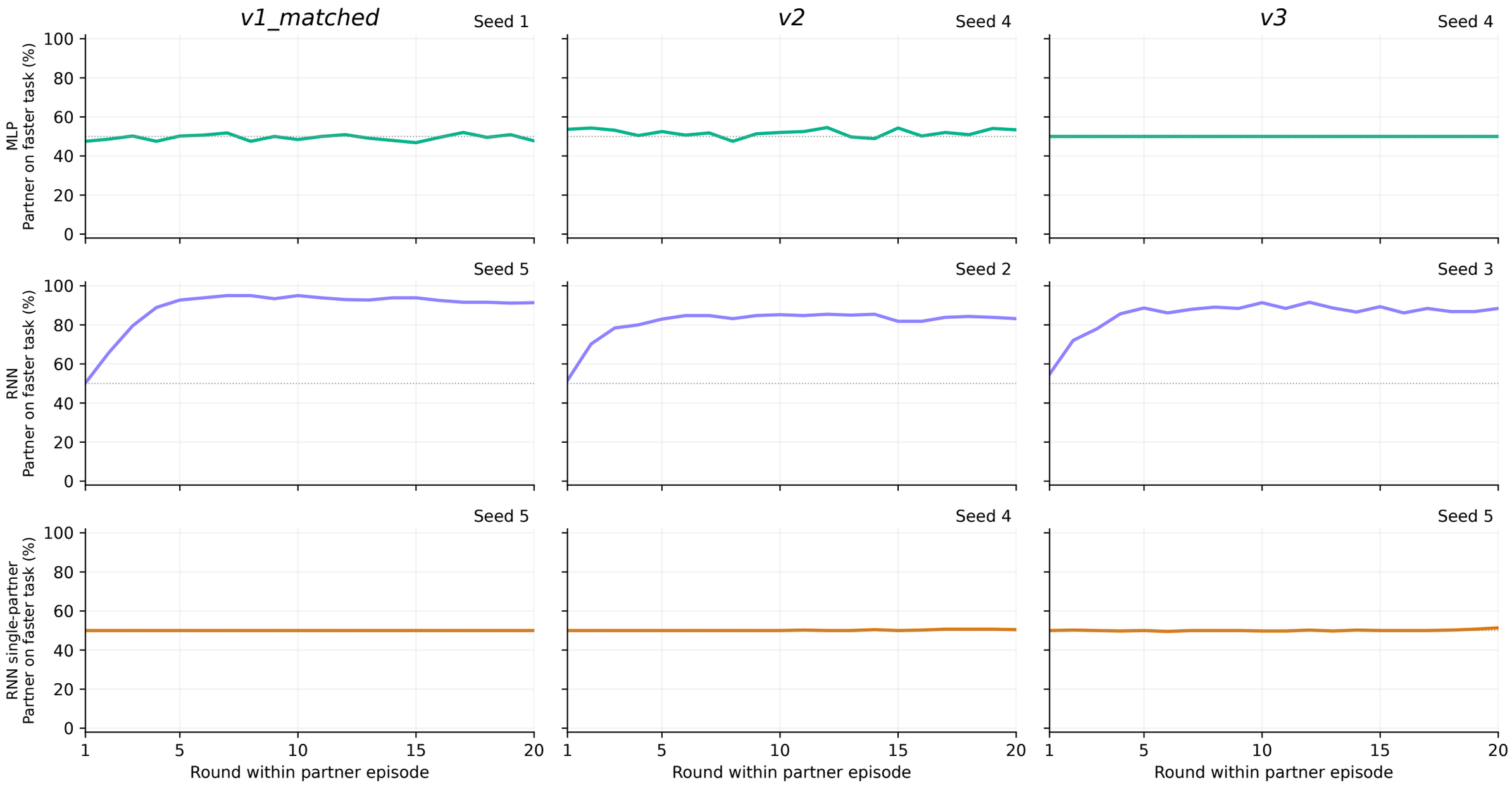
Final red and blue assignments of the selected training seeds

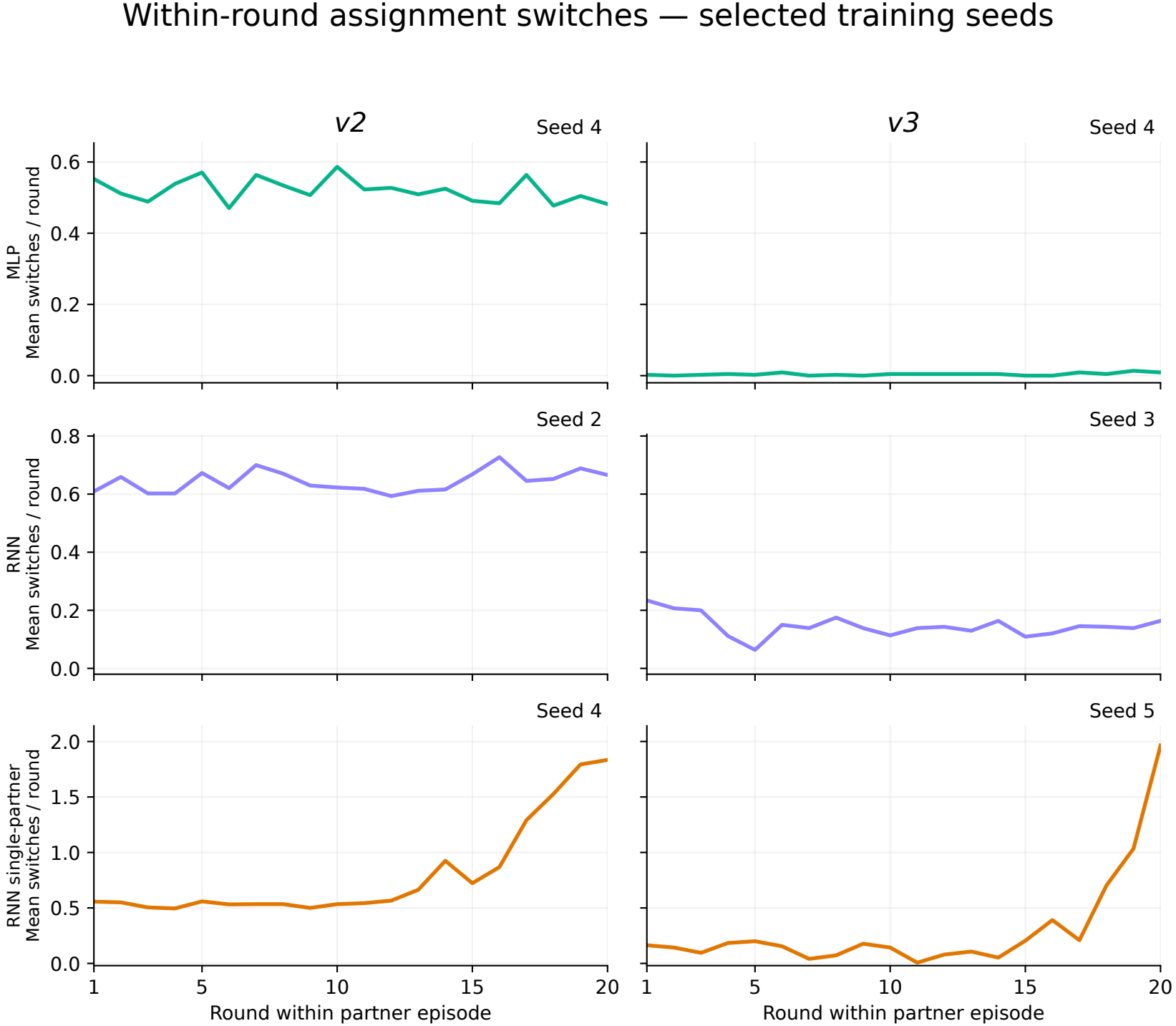


Same policy selections as the best-of-five held-out return figure; selected by final logged training return.
Red = ego red / partner blue; blue = ego blue / partner red. Final assignments average 440 episodes at each round.

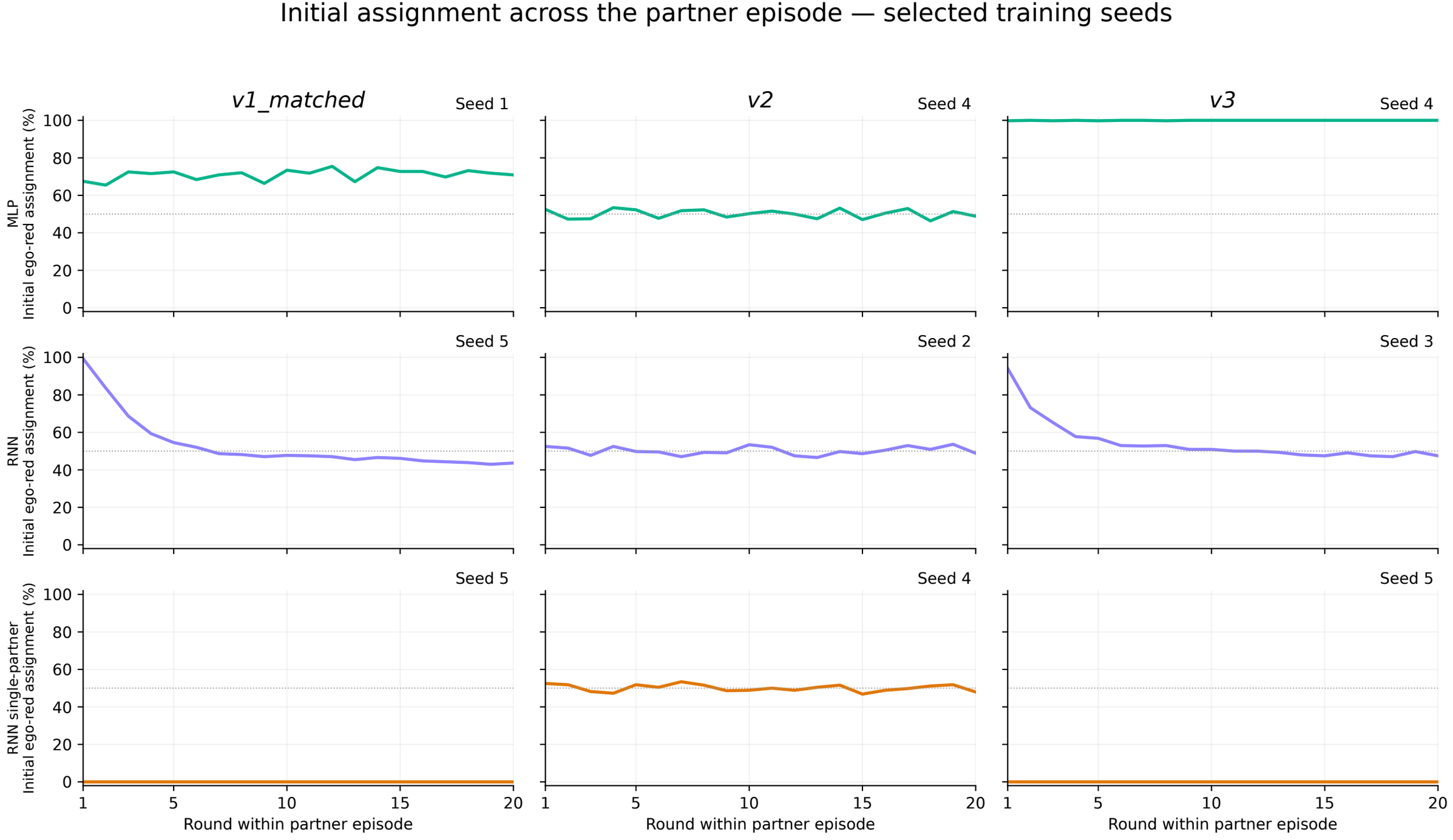

Final assignment across the partner episode — selected training seeds
v1_matched
v2
v3
Seed 1
Seed 4
Seed 4
Seed 5
Seed 2
Seed 3
Seed 5
Seed 4
Seed 5
MLP
RNN
RNN single-partner
Final ego-red assignment (%)
0
20
40
60
80
100
1
5
10
15
20
Round within partner episode
One policy per experiment and condition, selected by final logged training return among seeds 1–5.
Each curve averages 440 held-out episodes. Seed choices match the best-of-five return figure.

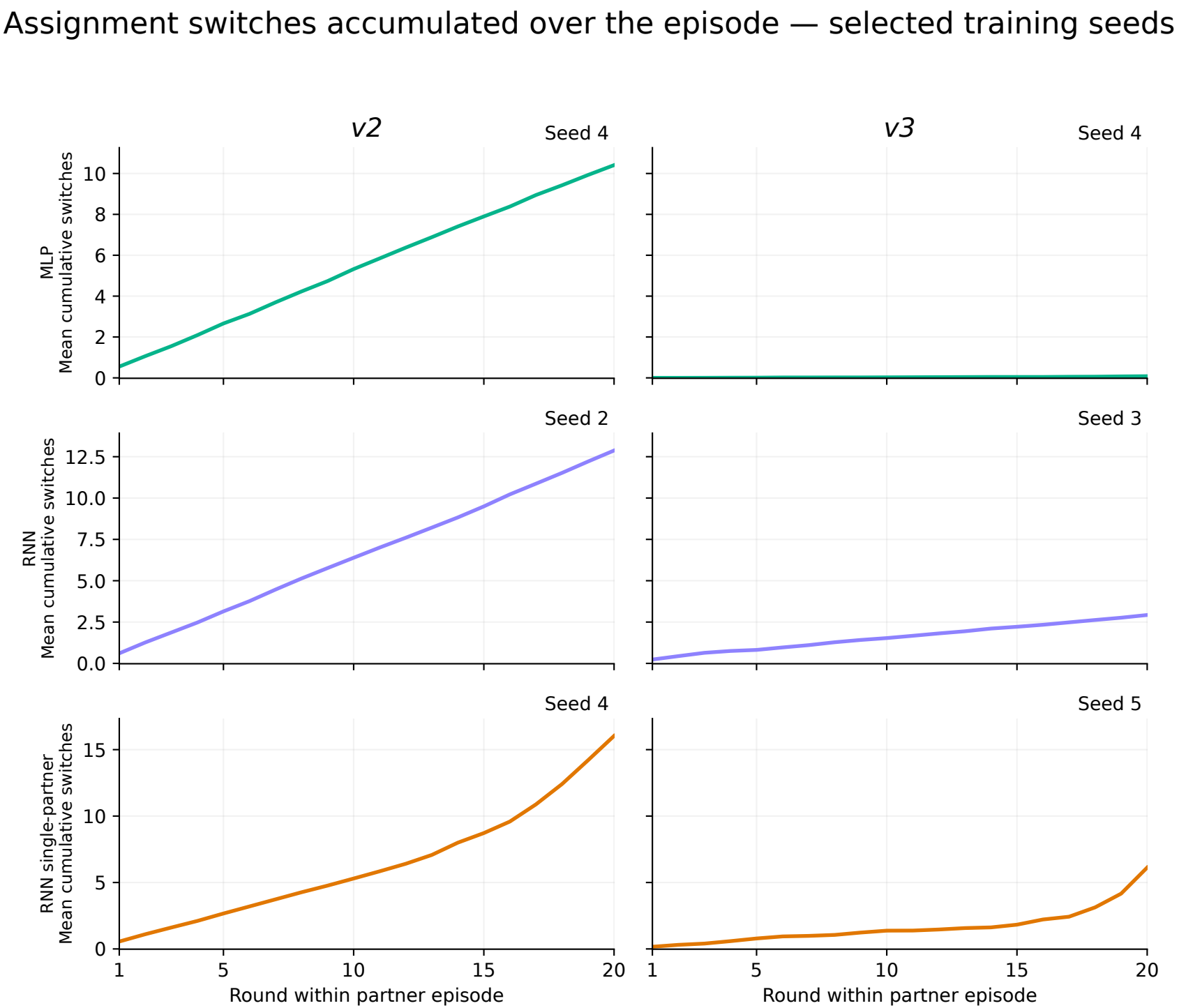

Final blue assignment across the partner episode — selected training seeds
v1_matched
v2
v3
Seed 1
Seed 4
Seed 4
Seed 5
Seed 2
Seed 3
Seed 5
Seed 4
Seed 5
MLP
RNN
RNN single-partner
Final ego-blue assignment (%)
0
20
40
60
80
100
1
5
10
15
20
Round within partner episode
One policy per experiment and condition, selected by final logged training return among seeds 1–5.
Each curve averages 440 held-out episodes. Seed choices match the best-of-five return figure.


Final assignment matches partner's faster task — selected training seeds
v1_matched
v2
v3
Seed 1
Seed 4
Seed 4
Seed 5
Seed 2
Seed 3
Seed 5
Seed 4
Seed 5
MLP
RNN
RNN single-partner
Partner on faster task (%)
0
20
40
60
80
100
1
5
10
15
20
Round within partner episode
One policy per experiment and condition, selected by final logged training return among seeds 1–5.
Each curve averages 440 held-out episodes. Seed choices match the best-of-five return figure.



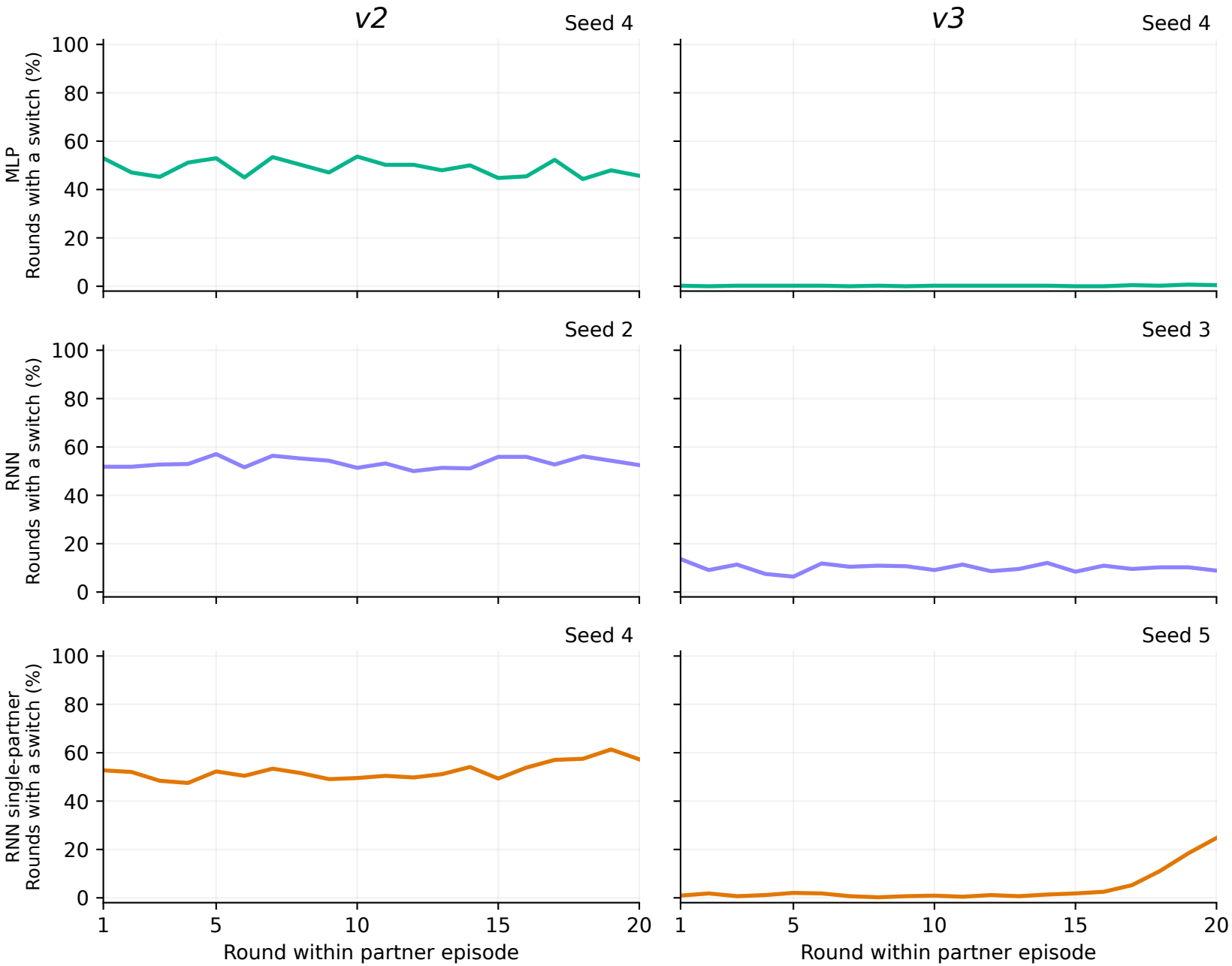
One policy per experiment and condition, selected by final logged training return among seeds 1–5.
Each curve averages 440 held-out episodes. Seed choices match the best-of-five return figure.


Initial assignment across the partner episode — selected training seeds
v1_matched
v2
v3
Seed 1
Seed 4
Seed 4
Seed 5
Seed 2
Seed 3
Seed 5
Seed 4
Seed 5
MLP
RNN
RNN single-partner
Initial ego-red assignment (%)
0
20
40
60
80
100
1
5
10
15
20
Round within partner episode
One policy per experiment and condition, selected by final logged training return among seeds 1–5.
Each curve averages 440 held-out episodes. Seed choices match the best-of-five return figure.

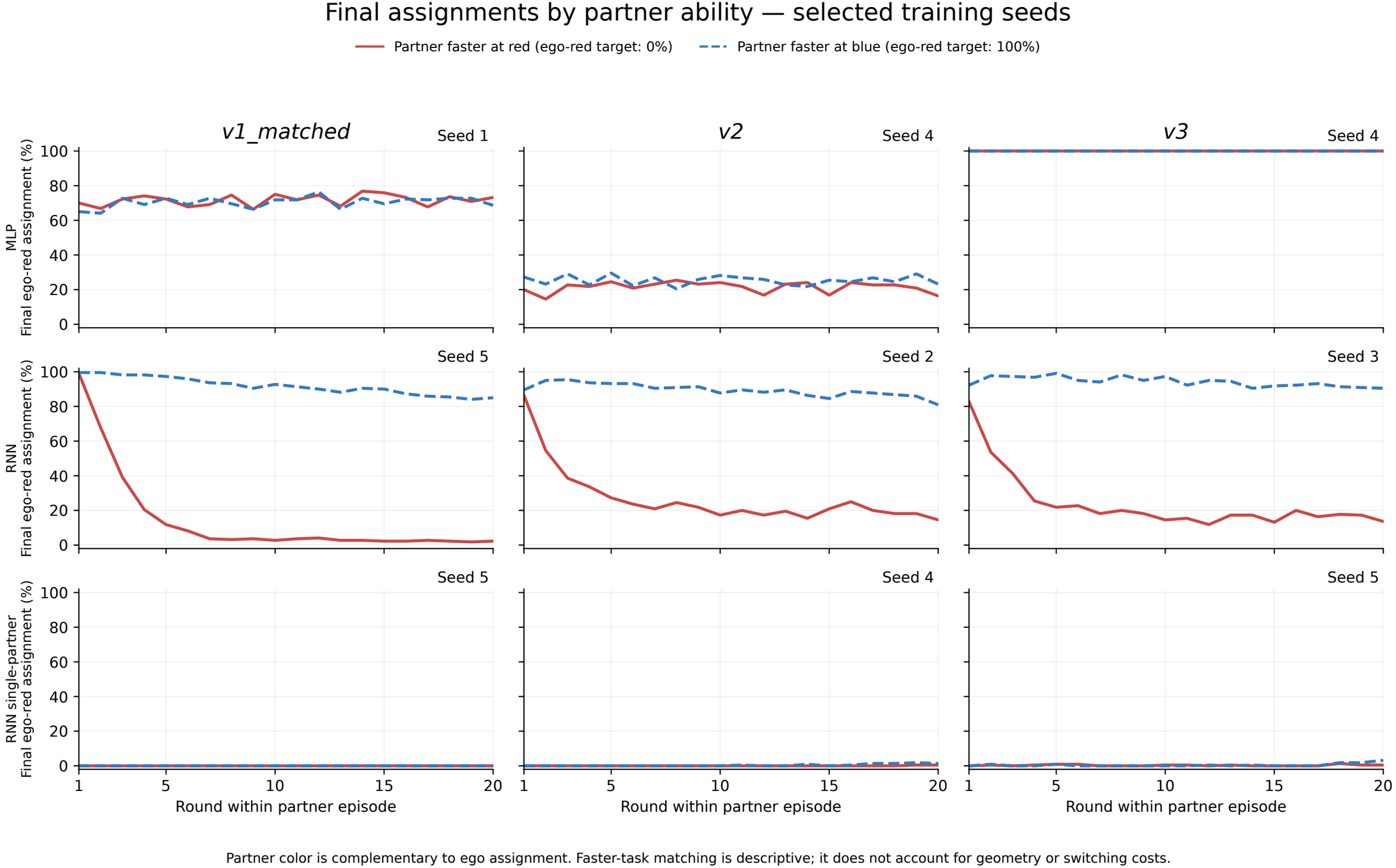


One policy per experiment and condition, selected by final logged training return among seeds 1–5.
Each curve averages 440 held-out episodes. Seed choices match the best-of-five return figure.

Rounds with at least one assignment switch — selected training seeds



One policy per experiment and condition, selected by final logged training return among seeds 1–5.
Each curve averages 440 held-out episodes. Seed choices match the best-of-five return figure.


Final assignments by partner ability — selected training seeds
Partner faster at red (ego-red target: 0%)
Partner faster at blue (ego-red target: 100%)
v1_matched
v2
v3
Seed 1
Seed 4
Seed 4
Seed 5
Seed 2
Seed 3
Seed 5
Seed 4
Seed 5
MLP
RNN
RNN single-partner
Final ego-red assignment (%)
0
20
40
60
80
100
1
5
10
15
20
Round within partner episode
Partner color is complementary to ego assignment. Faster-task matching is descriptive; it does not account for geometry or switching costs.